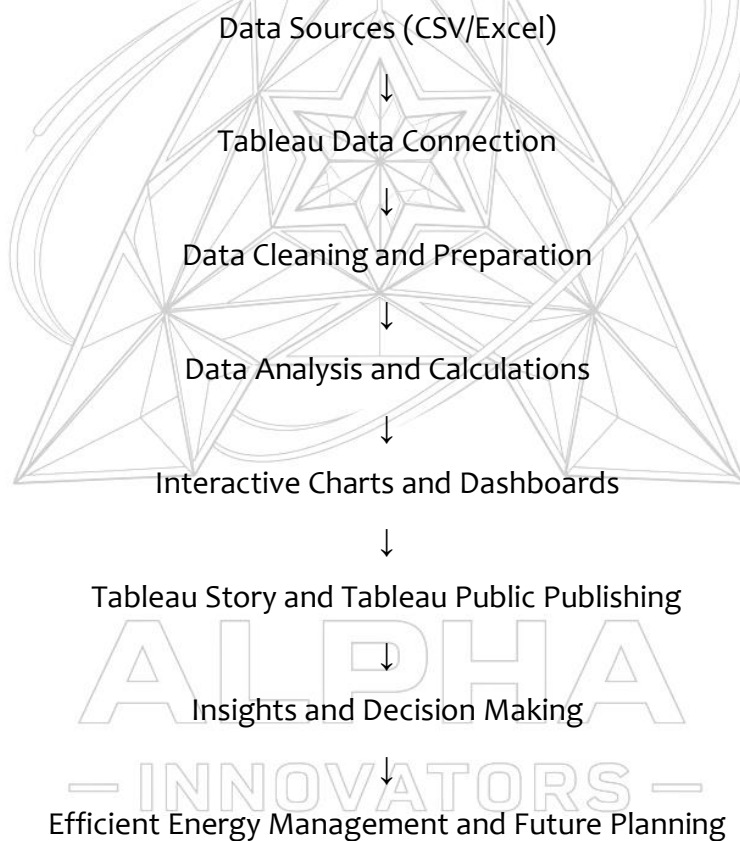


## Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

|               |                                                                                            |
|---------------|--------------------------------------------------------------------------------------------|
| TEAM ID       | SWTID-2026-2268                                                                            |
| PROJECT TITLE | Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau |
| MAX MARKS     | 5 MARKS                                                                                    |

### 4.3 – Solution Architecture

#### Architecture Flow



#### 1. Data Collection

- Gather electricity consumption data from publicly available datasets and other reliable sources.
- Store the data in CSV or Excel format.

#### 2. Data Connectivity

# Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

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- Connect the dataset to Tableau for analysis and visualization.
- Ensure proper data import and integration.

## 3. Data Preparation

- Clean and preprocess the data by handling missing values, removing duplicates, and formatting fields.
- Create calculated fields and perform necessary transformations.

## 4. Data Analysis

- Analyse electricity consumption trends based on years, regions, and other relevant factors.
- Identify peak usage periods and consumption patterns.

## 5. Dashboard Development

- Create interactive visualizations such as bar charts, line charts, maps, and trend analyses.
- Develop dashboards and stories to present insights effectively.

## 6. Publishing and Sharing

- Publish the dashboards to Tableau Public.
- Enable users to access and interact with the visualizations online.

## 7. Decision Support

- Support efficient energy planning and sustainable electricity usage.

ALPHA  
— INNOVATORS —  
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